

A Coordinate-Safe Phase Portrait for the Bose–Josephson Dimer

HCS Research Program

30 August 2026 (revision 1)

Abstract

We study the normalized two-mode Bose–Josephson dimer $H(z, \phi) = \Lambda z^2/2 - \sqrt{1 - z^2} \cos \phi$, $\Lambda \geq 0$. Passing to the Bloch sphere removes the apparent coordinate singularity at $z = \pm 1$. We classify $(0, 0)$, $(0, \pi)$, and the $\Lambda > 1$ symmetry-broken equilibria, including the $\Lambda = 1$ pitchfork. Elimination of the phase gives an exact quartic in z ; its roots yield complete-elliptic- K periods on crossing and self-trapped components. The $H = 1$ sech homoclinic, its $\Lambda = 2$ pole limit, and the reverse component criterion are stated without assigning a finite period to a separatrix. This is a source-local phase-portrait theorem: no arithmetic owner or target determinant is claimed.

1 Hamiltonian and Bloch regularization

Set

$$H(z, \phi) = \frac{\Lambda z^2}{2} - \sqrt{1 - z^2} \cos \phi, \quad \Lambda \geq 0,$$

with canonical equations

$$\dot{z} = -\sqrt{1 - z^2} \sin \phi, \quad \dot{\phi} = \Lambda z + \frac{z \cos \phi}{\sqrt{1 - z^2}}. \quad (1)$$

The Bloch variables $x = \sqrt{1 - z^2} \cos \phi$, $y = \sqrt{1 - z^2} \sin \phi$ satisfy

$$\dot{x} = -\Lambda z y, \quad \dot{y} = z(1 + \Lambda x), \quad \dot{z} = -y. \quad (2)$$

Thus the north and south poles have $\dot{y} = +1$ and -1 , respectively; they are not equilibria and no value of the singular angle ϕ is assigned there.

Theorem 1 (Fixed points and pitchfork). *The points $(0, 0)$ and $(0, \pi)$ exist for every Λ . The first has energy -1 , linearization $\begin{bmatrix} 0 & -1 \\ \Lambda+1 & 0 \end{bmatrix}$, and frequency $\sqrt{\Lambda+1}$. The second has energy $+1$ and linearization $\begin{bmatrix} 0 & 1 \\ \Lambda-1 & 0 \end{bmatrix}$: it is elliptic for $\Lambda < 1$, parabolic at $\Lambda = 1$, and hyperbolic for $\Lambda > 1$. For $\Lambda > 1$, two additional elliptic points are*

$$z = \pm \sqrt{1 - \Lambda^{-2}}, \quad \phi = \pi, \quad H_{\max} = \frac{\Lambda + \Lambda^{-1}}{2}. \quad (3)$$

In the canonical $(z, \phi - \pi)$ coordinates their matrix is $\begin{bmatrix} 0 & 1/\Lambda \\ -\Lambda(\Lambda^2 - 1) & 0 \end{bmatrix}$, so their small-amplitude period is $2\pi/\sqrt{\Lambda^2 - 1}$. The zero-phase crossing limit is $2\pi/\sqrt{\Lambda + 1}$, and (3) coalesces with $(0, \pi)$ at the pitchfork $\Lambda = 1$.

Proof. Differentiate (1) and solve $\dot{z} = \dot{\phi} = 0$ away from the poles. The displayed matrices follow by first-order expansion. Equation (2) is obtained by the chain rule and extends the vector field over both poles. $\square$

2 Energy quartic and regular periods

On $H = h$,

$$z^2 = (1 - z^2) - \left(\frac{\Lambda z^2}{2} - h \right)^2 = -\frac{\Lambda^2}{4} z^4 + (\Lambda h - 1) z^2 + 1 - h^2. \quad (4)$$

For $\Lambda > 0$, let

$$y_{\pm} = \frac{2(\Lambda h - 1 \pm \sqrt{\Lambda^2 - 2\Lambda h + 1})}{\Lambda^2}. \quad (5)$$

Then $z^2 = (\Lambda^2/4)(y_+ - z^2)(z^2 - y_-)$. If $-1 < h < 1$, $y_- < 0 < y_+$, the level is one connected crossing component and

$$T_{\text{cross}} = \frac{8}{\Lambda \sqrt{y_+ - y_-}} K \left(\sqrt{\frac{y_+}{y_+ - y_-}} \right). \quad (6)$$

If $\Lambda > 1$ and $1 < h < H_{\max}$, there are two sign-preserving components and each has

$$T_{\text{self}} = \frac{4}{\Lambda\sqrt{y_+}} K\left(\sqrt{1 - \frac{y_-}{y_+}}\right). \quad (7)$$

Here K uses the modulus convention; the code passes its square to the numerical library. The limits near the two elliptic centers are exactly the frequencies stated above.

3 Separatrix and component criterion

For $\Lambda > 1, h = 1$, (4) has the homoclinic solution

$$z(t) = \pm \frac{2\sqrt{\Lambda - 1}}{\Lambda} \operatorname{sech}(\sqrt{\Lambda - 1} t). \quad (8)$$

The full critical level is connected through the saddle at $z = 0$, with two one-sided homoclinic branches. At its turning point the phase is π for $1 < \Lambda < 2$, the point is a Bloch pole at $\Lambda = 2$, and the phase is 0 for $\Lambda > 2$. Formula (8) has infinite period; it is not a regular periodic orbit.

For regular $\Lambda > 1$ levels, $1 < h < H_{\max}$ is exactly the self-trapped regime: the two connected components preserve the sign of z , and changing the initial sign selects the reflected component. For $-1 < h < 1$, the single component crosses $z = 0$, so both signs occur. At $h = 1$, sign change is only asymptotic. At $\Lambda = 1, h = 1$, (4) reduces to $-z^4/4$, so only the isolated degenerate point $z = 0$ remains. At $\Lambda = 0$, $H = -x$ gives rigid Bloch rotation of period 2π on regular circles.

4 Certified finite receipt

The accompanying receipt contains 14 fixed-point rows, 8 pole rows, 13 level rows, and 5 component criteria. A producer-independent checker makes 995 assertions. SymPy verifies the Hamilton/Bloch identities, quartic roots, pitchfork and sech profile, while three independent transformed quadratures reproduce (6)–(7). Byte replay passes and a hostile suite rejects 28/28 repaired-hash mutations.

References

- [1] A. Smerzi, S. Fantoni, S. Giovanazzi and S. Shenoy, Quantum coherent atomic tunneling between two trapped Bose–Einstein condensates, *Phys. Rev. Lett.* 79 (1997), 4950–4953, DOI: 10.1103/PhysRevLett.79.4950, <https://doi.org/10.1103/PhysRevLett.79.4950>.
- [2] S. Raghavan, A. Smerzi, S. Fantoni and S. Shenoy, Coherent oscillations between two weakly coupled Bose–Einstein condensates, *Phys. Rev. A* 59 (1999), 620–633, DOI: 10.1103/PhysRevA.59.620, <https://doi.org/10.1103/PhysRevA.59.620>.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic, target-zero, determinant-matching, or Hilbert–Pólya claim. **Data and code.** All rows and audits accompany HCS-C243. This is not external peer review. **AI-use disclosure.** Generative tools assisted drafting and code generation; the artifact chain checks displayed formulas and metadata.